

Yusuf Nilsson

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EDUCATION

- University of Waterloo** | *Bachelor of Computer Science (Co-op)* 2021 - 2025
- Achieved Dean's List honors with 3.8 GPA in core systems and database courses
 - Completed 3 industrial co-op terms at top-tier tech companies, gaining hands-on backend development experience
- Queen Mary University of London** | *Summer Exchange - Distributed Systems Specialization* Summer 2023
- Studied advanced microservices architecture and cloud infrastructure patterns
 - Built 2 production-grade distributed system prototypes in Go and Rust

EXPERIENCE

- Stripe** | *Backend Engineering Intern* Jan 2024 - Apr 2024
- Optimized payment reconciliation pipeline to reduce processing latency by 38%, improving real-time reporting accuracy for 2,000+ merchants
 - Designed and deployed 3 new REST API endpoints for subscription management, handling 5M+ daily requests with 99.95% uptime
 - Reduced database query execution time by 42% through strategic indexing and query optimization on PostgreSQL
- Notion** | *Junior Backend Developer* May 2023 - Aug 2023
- Built real-time synchronization service using WebSockets and event streaming, decreasing client-server sync latency from 450ms to 120ms
 - Implemented Redis caching layer for frequently accessed database queries, reducing backend CPU usage by 31% and cutting response times by 55%
 - Mentored 2 interns on TypeScript best practices and helped them ship 4 feature flags that improved product reliability
- Datadog** | *Software Engineering Co-op* Sep 2022 - Dec 2022
- Engineered high-throughput log ingestion service in Go that processed 120M+ events daily with sub-100ms p99 latency
 - Debugged and fixed 12 production incidents related to metric aggregation, improving data consistency by 26%
 - Developed comprehensive integration tests that increased code coverage from 67% to 84% and caught 8 critical edge cases

PROJECTS

- AsyncCache** | *Go, Redis, gRPC, Docker* Feb 2024 - Apr 2024
- Built a distributed caching service supporting 50K+ concurrent connections with automatic invalidation, achieving 94% cache hit rate
 - Implemented cache coherence protocol reducing stale data scenarios by 73% across multi-node deployments
- PulseDB** | *Rust, PostgreSQL, Kafka, Tokio* Nov 2023 - Jan 2024
- Created time-series database engine handling 1M+ data points per second with 99.9% accuracy
 - Decreased memory footprint by 48% through columnar storage optimization and compression algorithms
- ShardFlow** | *Python, FastAPI, SQLAlchemy, Prometheus* Jul 2023 - Sep 2023
- Developed database sharding middleware that horizontally scaled read operations by 340% without application refactoring
 - Reduced query routing latency from 8ms to 1.2ms through consistent hashing optimization
- MicroTrace** | *TypeScript, Node.js, Jaeger, OpenTelemetry* Apr 2023 - Jun 2023
- Engineered distributed tracing library for microservices, enabling engineers to identify performance bottlenecks 65% faster
 - Reduced trace processing overhead from 12% to 2% of request latency through intelligent sampling
- EventStorm** | *Java, Spring Boot, Kafka, Elasticsearch* Dec 2022 - Feb 2023
- Built event sourcing framework processing 200K+ domain events hourly with zero data loss guarantees
 - Achieved 99.99% event delivery reliability through idempotency tokens and retry logic

EXTRACURRICULAR

- Waterloo Distributed Systems Club** | *Core Contributor & Workshop Lead* 2022 - Present
- Organized and led 8 technical workshops on Kafka, gRPC, and consensus algorithms for 150+ students
 - Grew community from 40 to 280+ active members through curated content and mentorship programs
- Open Source Maintainer** | *Creator of go-fastcache library* 2023 - Present
- Published Go caching library with 2,400+ GitHub stars and 180+ community contributions
 - Reduced download bandwidth costs for users by 67% through compression innovations featured in Hacker News top 5

TECHNICAL SKILLS

Languages – Go, Python, TypeScript, Rust, Java, SQL

Technologies – PostgreSQL, Redis, Kafka, gRPC, REST APIs, GraphQL, Docker, Kubernetes, Apache Spark

Tools – Git, GitHub, Datadog, New Relic, Prometheus, Grafana, CI/CD pipelines, Linux, VS Code